

Universal Linear Algebra Compliance Audit (Mode: NORMAL) (ALL)

Audit Context:
Mode: NORMAL
Reference: Episteme (Standard)
Matrix Size: 12

Provider	Operation	Domain	Result Summary
Generic JVM	Apache Commons Math (Optimized)	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
Generic JVM	Colt (Optimized)	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
Generic JVM	EJML (Optimized)	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
CPU (CARMA)	Episteme (CARMA)	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
CPU (Standard)	Episteme (Standard)	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
CPU (Strassen)	Episteme (Strassen)	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
CPU (Standard JVM)	Episteme CPU (Dense)	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
Generic JVM	Episteme CPU (Sparse)	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
Generic JVM	Episteme CPU Foundation	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
Generic JVM	Episteme Distributed Linear Algebra Provider (LocalDistributedContext)	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
Generic JVM	Episteme gRPC Remote (localhost:50051)	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected

Generic JVM	JBlas (Optimized)	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
CPU (Native ND4J)	ND4J (Native-Dense)	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
CPU (Native/Panama)	Native CPU-BLAS (Complex)	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
CPU (Native/Panama)	Native CPU-BLAS (Real)	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
N/A	Native CUDA Dense Linear Algebra Backend	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
N/A	Native CUDA Sparse Linear Algebra Backend	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
CPU (FFM-BLAS)	Native FFM-BLAS (Complex)	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
CPU (FFM-BLAS)	Native FFM-BLAS (Real)	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
CPU (Panama/MPFR)	Native MPFR Dense Linear Algebra Backend	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
Generic JVM	Native MPFR Sparse Linear Algebra Backend	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
Generic JVM	Native OpenCL Dense Linear Algebra Backend	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
Generic JVM	Native OpenCL Sparse Linear Algebra Backend	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
Generic JVM	Native SIMD (Complex)	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected
Generic JVM	Native SIMD (Real)	Linear Algebra Compliance	FAST/NORMAL/EXACT results collected

Operation Comparison: Apache Commons Math (Optimized) (ALL)

Episteme Multimodal Audit | Apache Commons Math (Optimized)

Operation Comparison: Colt (Optimized) (ALL)

Episteme Multimodal Audit | Colt (Optimized)

Operation Comparison: EJML (Optimized) (ALL)

Episteme Multimodal Audit | EJML (Optimized)

Operation Comparison: Episteme (CARMA) (ALL)

Episteme Multimodal Audit | Episteme (CARMA)

Operation Comparison: Episteme (Standard) (ALL)

Episteme Multimodal Audit | Episteme (Standard)

Operation Comparison: Episteme (Strassen) (ALL)

Episteme Multimodal Audit | Episteme (Strassen)

Operation Comparison: Episteme CPU (Dense) (ALL)

Episteme Multimodal Audit | Episteme CPU (Dense)

Operation Comparison: Episteme CPU (Sparse) (ALL)

Episteme Multimodal Audit | Episteme CPU (Sparse)

Operation Comparison: Episteme CPU Foundation (ALL)

Episteme Multimodal Audit | Episteme CPU Foundation

Operation Comparison: Episteme Distributed Linear Algebra Provider (LocalDistributedContext) (ALL)

Episteme Multimodal Audit | Episteme Distributed Linear Algebra Provider (LocalDistributedContext)

Operation Comparison: Episteme gRPC Remote (localhost:50051) (ALL)

Episteme Multimodal Audit | Episteme gRPC Remote (localhost:50051)

Operation Comparison: JBlas (Optimized) (ALL)

Episteme Multimodal Audit | JBlas (Optimized)

Operation Comparison: ND4J (Native-Dense) (ALL)

Episteme Multimodal Audit | ND4J (Native-Dense)

Operation Comparison: Native CPU-BLAS (Complex) (ALL)

Episteme Multimodal Audit | Native CPU-BLAS (Complex)

Operation Comparison: Native CPU-BLAS (Real) (ALL)

Episteme Multimodal Audit | Native CPU-BLAS (Real)

Operation Comparison: Native CUDA Dense Linear Algebra Backend (ALL)

Episteme Multimodal Audit | Native CUDA Dense Linear Algebra Backend

Operation Comparison: Native CUDA Sparse Linear Algebra Backend (ALL)

Episteme Multimodal Audit | Native CUDA Sparse Linear Algebra Backend

Operation Comparison: Native FFM-BLAS (Complex) (ALL)

Episteme Multimodal Audit | Native FFM-BLAS (Complex)

Operation Comparison: Native FFM-BLAS (Real) (ALL)

Episteme Multimodal Audit | Native FFM-BLAS (Real)

Operation Comparison: Native MPFR Dense Linear Algebra Backend (ALL)

Episteme Multimodal Audit | Native MPFR Dense Linear Algebra Backend

Operation Comparison: Native MPFR Sparse Linear Algebra Backend (ALL)

Episteme Multimodal Audit | Native MPFR Sparse Linear Algebra Backend

Operation Comparison: Native OpenCL Dense Linear Algebra Backend (ALL)

Episteme Multimodal Audit | Native OpenCL Dense Linear Algebra Backend

Operation Comparison: Native OpenCL Sparse Linear Algebra Backend (ALL)

Episteme Multimodal Audit | Native OpenCL Sparse Linear Algebra Backend

Operation Comparison: Native SIMD (Complex) (ALL)

Episteme Multimodal Audit | Native SIMD (Complex)

Operation Comparison: Native SIMD (Real) (ALL)

Episteme Multimodal Audit | Native SIMD (Real)